Jessica Thompson

Computational Cognitive Scientist

Contact

jessica.thompson@psy.ox.ac.uk
http://thompsonj.github.io

Research Interests

Learning, cognitive science, human-AI interaction, AI alignment, collective intelligence, metascience, cooperation, generalization, understanding deep learning, philosophy of science

Core Competencies

Problem formulation, critical thinking, multi-disciplinary synthesis, open-mindedness

Computer Skills

Proficient with Python, MATLAB, Java, shell scripting, LaTeX

Links

G Scholar: TJweLP0AAAAJ
Github: thompsonj
LinkedIn: jessica-af-thompson

Languages

English (native) and French (professional fluency)

Education

PhD Cognitive Science and Neuropsychology — Université de Montréal **2013–2020**
Supervised by Marc Schönwiesner and Yoshua Bengio

MA Digital Musics — Dartmouth College **2011–2013**
Supervised by Michael Casey

BA Psychology (Hons.), Computer Science, Music Technology — McGill University **2006–2011**
Supervised by Stephen McAdams, Evan Balaban and Robert Zatorre

Research Experience

School of Psychology and Neuroscience, University of Glasgow — Glasgow, Scotland **Starting Dec 2026**
Lecturer
Lead independent research on human and artificial cognition. Supervise research students and contribute to the delivery of teaching at all levels.

Human Information Processing Lab, University of Oxford — Oxford, United Kingdom **March 2021–Nov 2026**
Postdoctoral Researcher
Use artificial neural networks to model and improve human learning and reasoning. Lead subgroup on curriculum learning and behavioural cloning.

Quebec Artificial Intelligence Institute (Mila) — Montréal, Canada **2014–2020**
PhD Researcher
Explored methods to compare deep networks to measurements of human neural activity

International Laboratory for Research on Brain, Music & Sound — Montréal, Canada **2013–2020**
PhD Researcher
Designed and conducted study on the representation of spectro-temporal modulations using machine learning-based analysis of 7T fMRI responses to dynamic ripples

Nuance Communications Canada Inc. — Montréal, Canada **2015–2018**
Mitacs PhD Fellow
Analyzed the transferability of intermediate features across languages in convolutional neural network-based acoustic models in automatic speech recognition systems

Department of Cognitive Neuroscience, Maastricht University — Maastricht, The Netherlands **2015–2016**
Visiting Researcher
Designed and conducted 7T fMRI study to model responses to native and non-native speech quilts with convolutional neural networks trained on natural speech

Bregman Media Labs, Dartmouth College — Hanover, USA **2011–2013**
Graduate Researcher
Conducted fMRI and EEG experiments about neural decoding of various acoustic and semantic music features

Distributed Digital Music Archives and Libraries Lab, McGill University — Montréal, Canada **2008–2011**
Programmer
Developed software in Java and Python for machine learning and information retrieval services with application to music research

Auditory Cognitive Neuroscience Lab, Montreal Neurological Institute — Montréal, Canada **2010–2011**
Student Researcher
Assessed the effect of circularity (non-independence) in whole brain fMRI region-of-interest (ROI) analysis

MEG Lab, Rotman Research Institute — Toronto, Canada **2010**
Student Researcher
Conducted MEG and behavioural experiments on binaural auditory beating

Music Perception and Cognition Lab, McGill University — Montréal, Canada **2008–2009**
Data Analyst
Employed continuous data analysis techniques to analyse psychophysiological signals

Publications

Highlighted publications

William D'Alessandro and **Jessica Thompson**. "AI Surrogacy in Psychological Research". In: The Role of Artificial Intelligence in Science: Methodological and Epistemological Studies. Routledge, in press.

Reward models inherit value biases from pretraining
Brian Christian, **Jessica Thompson**, *Elle Michelle Yang, Vincent Adam, Hannah Rose Kirk, Christopher Summerfield, and Tsvetomira Dumbalska*
International Conference on Learning Representations (ICLR) (2026)

Reward Model Interpretability Via Optimal and Pessimal Tokens
Brian Christian, Hannah Rose Kirk, **Jessica Thompson**, *Christopher Summerfield*[*], *and Tsvetomira Dumbalska*[*]
ACM Conference on Fairness, Accountability & Transparency (FAccT) (2025)

Journal articles

Humans and neural networks show similar patterns of transfer and interference during continual learning
Eleanor Holton, Lukas Braun, **Jessica Thompson**, *Jan Grohn, and Christopher Summerfield*
Nature Human Behaviour. 2026

Zero-shot counting with a dual-stream neural network model
Jessica Thompson, *Hannah Sheahan, Tsvetomira Dumbalska, Julian Sandbrink, Manuela Piazza, and Christopher Summerfield*
Neuron. 112: 24 4147–4158.e5. 2024

Thinking beyond the ventral stream: Commentary on Bowers et al.
Christopher Summerfield and **Jessica Thompson**
Behavioral and Brain Sciences. e409: 46. 2023

Forms of explanation and understanding for neuroscience and artificial intelligence
Jessica Thompson
Journal of Neurophysiology. 126 pp. 1860–1874. 2021

Primer: Noise increases the correspondence between artificial and human vision
Jessica Thompson
PLoS Biology. 19: 13 pp. 4–7. 2021

Training neural networks to recognize speech increased their correspondence to the human auditory pathway but did not yield a shared hierarchy of acoustic features
Jessica Thompson, *Federico De Martino, Yoshua Bengio, Elia Formisano, and Marc Schönwiesner*
bioRxiv preprint. 2021

Human cortical responses to slow and fast binaural beats reveal multiple mechanisms of binaural hearing
Bernhard Ross, Takahiro Miyazaki, **Jessica Thompson**, *Shahab Jamali, and Takako Fujioka*
Journal of Neurophysiology. 112.8 pp. 1871–1884. 2014

Sound envelope encoding in the auditory cortex revealed by neuromagnetic responses in the theta to gamma frequency bands
Takahiro Miyazaki, **Jessica Thompson**, *Takako Fujioka, and Bernhard Ross*
Brain Research. 1506 pp. 64–75. 2013

Conference and workshop proceedings

Learning to count visual objects by combining "what" and "where" in recurrent memory
Jessica Thompson, *Hannah Sheahan, and Christopher Summerfield*
Proceedings of the 1st Gaze Meets ML workshop, PMLR. Vol. 210 (2023) pp. 199–218

How transferable are features in convolutional neural network acoustic models across languages?
Jessica Thompson, *Marc Schönwiesner, Yoshua Bengio, and Daniel Willett*
Proceedings of the IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP) (2019) pp. 2827–2831

Conference ticket allocation via non-uniform random selection to address systemic biases
Jessica Thompson, *Laurent Dinh, Nicolas Le Roux, and Layla El Asri*
Neural Information Processing Systems (NeurIPS) workshop on Correcting and Critiquing Trends in Machine Learning (2018)

How can deep learning advance computational modeling of sensory information processing?
Jessica Thompson, *Yoshua Bengio, Elia Formisano, and Marc Schönwiesner*
Neural Information Processing Systems workshop on Representation Learning in Artificial and Biological Neural Networks (MLINI) (2016)

Audio stimulus reconstruction using multi-source semantic embedding
Jessica Thompson, *Michael Casey, and Lorenzo Torresani*
Neural Information Processing Systems workshop on Machine Learning and Interpretation in Neuroimaging (MLINI) (Dec. 2013)

Digital document image retrieval using optical music recognition
Andrew Hankinson, John Ashley Burgoyne, Gabriel Vigliensoni, Alastair Porter, **Jessica Thompson**, *Wendy Liu, Remi Chiu, and Ichiro Fujinaga*
Proceedings of the 13th Intl. Society for Music Information Retrieval Conference (ISMIR) (Oct. 2012) pp. 577–582

Musical neurosemantic decoding using online weighted approximate-rank pairwise loss optimization in a joint semantic space
Jessica Thompson, *Michael Casey, and Lorenzo Torresani*
Neural Information Processing Systems workshop on Machine Learning and Interpretation in Neuroimaging (MLINI) (Dec. 2012). Lake Tahoe, USA

Music imagery information retrieval: Bringing the song on your mind back to your ears
Sebastian Stober and **Jessica Thompson**
Proceedings of the 13th International Conference on Music Information Retrieval (ISMIR) (Oct. 2012)

Population codes representing musical timbre for high-level fMRI categorization of music genres
Michael Casey, **Jessica Thompson**, *Olivia Kang, and Thalia Wheatley*
Neural Information Processing Systems workshop on Machine Learning and Interpretation in NeuroImaging (MLINI) (Dec. 2011)

Using the ACE XML 2.0 file formats to store and share music classification data
Cory McKay, John Ashley Burgoyne, **Jessica Thompson**, *and Ichiro Fujinaga*
Proceedings of the 10th International Conference on Music Information Retrieval (ISMIR) (Oct. 2009) pp. 303–308

Additions and improvements to the ACE 2.0 music classifier
Jessica Thompson, *Cory McKay, John Ashley Burgoyne, and Ichiro Fujinaga*
Proceedings of the 10th International Conference on Music Information Retrieval (ISMIR) (Oct. 2009) pp. 435–440

Selected Conference Abstracts and Presentations

Using behavioural cloning to accelerate human language learning
Jessica Thompson*, *Jirko Rubruck**, *Mohammed Amara, Vincent Adam, Nicholas Marsh, and Christopher Summerfield*
Behavioral Clones Workshop, 2026, Berlin, Germany

Zero-Shot Visual Numerical Reasoning with Dual-Stream Neural Networks
Jessica Thompson, *Hannah Sheahan, Tsvetomira Dumbalska, Julian Sandbrink, and Christopher Summerfield*
Computational and Systems Neuroscience (COSYNE) conference, 2024, Lisbon, Portugal

Humans and Neural Networks Show Similar Patterns of Transfer and Interference in a Continual Learning Task
Eleanor Holton, Lukas Braun, **Jessica Thompson**, *and Christopher Summerfield*
Cognitive Computational Neuroscience Conference, 2023, Oxford, UK

Zero-Shot Visual Numerical Reasoning with Dual-Stream Neural Networks
Jessica Thompson, *Hannah Sheahan, and Christopher Summerfield*
Cognitive Computational Neuroscience Conference, 2023, Oxford, UK

Making Sense of Probabilistic Representation
Samuel Lippl, **Jessica Thompson**, *Raphael Gerraty, and Gerardo Viera*
Society for Philosophy and Psychology, 2022, Milan, Italy

The effect of task and training on intermediate representations in convolutional neural networks revealed with modified RV similarity analysis
Jessica Thompson, *Marc Schönwiesner, and Yoshua Bengio*
Cognitive Computational Neuroscience (CCN) Conference, 2019, Berlin, Germany

Towards a common philosophy of explanation for artificial and biological intelligence
Jessica Thompson
Cognitive Computational Neuroscience (CCN) Conference, 2018, Philadelphia, USA, (Poster)

Encoding of dynamic ripple mixtures in human auditory cortex using 7T fMRI
Jessica Thompson, *Federico De Martino, Marc Schönwiesner, and Elia Formisano*
Annual Meeting of the Organization for Human Brain Mapping (OHBM), 2016, Geneva, Switzerland

Reconstructing musical audio features from continuous single-trial EEG
Jessica Thompson *and Michael Casey*

Annual Meeting of the Organization for Human Brain Mapping (OHBM), 2014, Hamburg, Germany

Experience, perception, and physicality in experimental music: An argument for the role of neuroscience in music phenomenology
Jessica Thompson
Cognitio - Creative Minds: Cognitive Sources of Art and Discovery, 2013, Montreal, Canada

Reconstructing musical audio features from continuous single-trial EEG
Jessica Thompson
Cognitively-Based Music Information Retrieval (CogMIR) workshop, 2013, Toronto, Canada

Music information retrieval from neurological signals: Towards neural population codes for music
Jessica Thompson *and Michael Casey*
Conference of the Society for Music Perception and Cognition (SMPC), 2013, Toronto, Canada

Predicting crowdsourced musical tags from brain activity
Jessica Thompson, *Michael Casey, and Lorenzo Torresani*
Cognitively Based Music Information Retrieval (CogMIR), 2012, Toronto, Canada

Searching the Liber Usualis: Using CouchDB and ElasticSearch to Query Graphical Music Documents
Jessica Thompson, *Andrew Hankinson, and Ichiro Fujinaga*
Proceedings of the 12th International Conference on Music Information Retrieval (ISMIR), 2011, Miami, USA

Left and right auditory cortices contribute differently to perception of slow and fast binaural beats
Takahiro Miyazaki, **Jessica Thompson**, *and Bernhard Ross*
Annual Meeting of the Cognitive Neuroscience Society (CNS), 2011, San Francisco, USA

Transition from Transient to Steady-State Gamma-Band Responses: An MEG Study on Acoustic Beats
Takahiro Miyazaki, **Jessica Thompson**, *and Bernhard Ross*
Annual Meeting of the Organization for Human Brain Mapping (OHBM), 2011, Québec, Canada

Teaching and Mentorship

Masters Supervisor Oxford, United Kingdom **2025–2026**
Department of Experimental Psychology, University of Oxford
Supervise M.Sci. student working on curriculum learning

Research Intern Supervisor Oxford, United Kingdom **Summer 2025**
Department of Experimental Psychology, University of Oxford
Supervised undergraduate internship about curriculum learning

Teaching Assistant Online **Summer 2024**
Neuromatch Academy - NeuroAI Program
Led problem-based learning tutorials on NeuroAI

Masters project supervisor Oxford, United Kingdom **Trinity 2023**
Department of Experimental Psychology, University of Oxford
Supervised human eyetracking experiment and neural network simulations

Undergraduate project supervisor Oxford, United Kingdom **Hilary 2023**
Department of Experimental Psychology, University of Oxford
Supervised human eyetracking and behavioural experiment

Tutor Oxford, United Kingdom **Michaelmas 2021**
How To Build A Brain From Scratch (Department of Experimental Psychology, University of Oxford)
Led tutorial discussion and provided feedback on essays.

Teaching Assistant Montréal, Canada **Winter 2020**
IFT 6135 - Representation Learning (Department of Computer Science, University of Montreal)
Prepared and graded practical and theoretical assignments for graduate level course on deep learning.

Mentor Online **2019-2020**
UNIQUE Student Symposium 2019 and Women in Machine Learning Workshop 2020
Led mentorship roundtables on surviving graduate school

Methods Workshop Organizer Montréal, Canada **Jun 2017**
Centre for Research on Brain, Language and Music
Designed and gave workshop on machine learning in python for psychologists and neuroscientists

Tutor/Supervisor Maastricht, The Netherlands **Spring 2015**
PSY2027: Research: How to do it? (Faculty of Psychology and Neuroscience, Maastricht University)
Supervised a group of 2nd year undergraduate students while they carried out all aspects of a research project about auditory perception

Media & Outreach

Covenant of Reciprocity: Review of Braiding Sweetgrass Science for the People Magazine **2024**
Book review in vol. 26, no. 2

Explanation and understanding in neuroscience and AI Journal of Neurophysiology podcast series **May 2022**
Podcast interview

The Deep Sociality of Science: Understanding Science as a Cooperative Process Science for the People Magazine **Jan 2022**
Book review in vol. 24, no. 3

Neuro-AI Explanation Brain Inspired **Jul 2021**
Podcast Interview

Invited Talks and Guest Lectures

How behavioural cloning has the potential to accelerate discovery in the behavioural sciences London, UK **Nov 2026**
Lecture Series on Human and Artificial Intelligence - London School of Economics

A cognitive science approach to AI alignment London, UK **Apr, May 2026**
Fundamentals of AI for civil servants - Imperial College London

Using AI to understand and improve human learning Toronto, Canada **Oct 2025**
Mini-Symposium on Cognition and Computation: Bidirectional Insights Between Cognitive Neuroscience and AI - Fields Institute

How AI is transforming how we understand the brain and mind Oxford, UK **May 2024**
Transformative Technologies Symposium - Somerville College

Zero-shot numerical reasoning in dual-stream recurrent neural networks and the primate visual system Oxford, UK **Feb 2024**
Oxford Neurotheory Forum

Tools and metrics are means, not ends Oxford, UK **Aug 2023**
Generative Adversarial Collaboration at the Cognitive Computational Neuroscience Conference

Explanation and understanding at the intersection of neuroscience and artificial intelligence Glasgow, Scotland **May 2023**
Philosophy, psychology and neuroscience seminars, University of Glasgow

Learning to count by learning to map: How spatial maps grounded in action support relational judgements Lisbon, Portugal **Mar 2022**
Computational and Systems Neuroscience (COSYNE) workshop on The neural codes of abstraction and the link to behavioral generalization

Forms of understanding and explanation for neuroscience and artificial intelligence Amsterdam, Netherlands **Nov 2021, 2022**
VU University of Amsterdam Honours Programme course concerning scientific theories

Comparing activations in artificial and biological neural networks Montevideo, Uruguay **Jul 2021**
IBRO-LARC Neuroscience and AI for all Virtual Associate School

Comparing activations in artificial and biological neural networks Montréal, Canada **Nov 2019, Dec 2020**
Montreal Artificial Intelligence and Neuroscience (MAIN) Conference

Theoretical motivations of deep learning as it relates to artificial intelligence and the brain Montréal, Canada **Jan 2017**
Breakout talk at the Canadian University Software Engineering Conference (CUSEC)

Funding & Awards

Postdoctoral Fellowship Canada Natural Sciences and Engineering Research Council (NSERC) **Dec 2023--Dec 2025**
National competition ($70,000 CAD/yr for two years)

Junior Research Fellowship Somerville College, Oxford **Dec 2023--Dec 2025**
Institutional competition

Departmental award for Public Policy Engagement Dept of Experimental Psychology, University of Oxford **April 2024**
Departmental competition

Travel Award Advances in NeuroAI workshop in Montreal **October 2023**
International competition ($1,500 CAD)

SSNAP Fellowship + Sub-award Summer Seminars in Neuroscience and Philosophy (SSNAP) (John Templeton Foundation) **2021-2022**
Co-PI ($30,000 USD)

Best Poster Award Montreal Artificial Intelligence and Neuroscience Conference **2019**
Local conference ($400)

Mitacs Accelerate PhD Fellowship Université de Montréal and Nuance Communications Inc. **2015–2018**
National competition ($30,000/yr for three years)

Best Poster Award Montreal Artificial Intelligence and Neuroscience Conference **2017**
Local conference ($400)

Fonds de recherche du Québec - Nature et technologies (FRQNT) Doctoral scholarship Université de Montréal **2015–2016**
Provincial competition ($26,666)

Erasmus Mundus Mobility Fellowship in Auditory Cognitive Neuroscience Maastricht University **2015**
International competition (€17,000)

Natural Sciences and Engineering Research Council (NSERC) CREATE Graduate Fellowship in Auditory Cognitive Neuroscience Université de Montréal **2014–2015**
National competition ($21,000)

Dartmouth Fellowship Dartmouth College **2011–2013**
Full tuition scholarship ($55,000/yr) and stipend ($20,000/yr) for two years

NSERC-CREATE Undergraduate Student Research Award in Auditory Cognitive Neuroscience Rotman Research Institute **2010**
National competition ($4,600)

NSERC-CREATE Undergraduate Student Research Award in Auditory Cognitive Neuroscience McGill University **2009**
National competition ($4,600)

Professional Service and Other Experience

Member of the Scholarships and Fellowships Review Committee for Computing Sciences Canada **2024-2027**
Natural Sciences and Engineering Research Council of Canada
Evaluate Scholarship and Fellowship applications

Satellite Event Co-Organizer Amsterdam, Netherlands **Aug 2025**
CCN satellite event on 'Science for Social Good'
Give opening remarks and coordinate event logistics for afternoon event

Membership & Recruitment Secretary, Member of the Executive Committee Oxford, UK **2023-Now**
University and College Union - University of Oxford Branch
Lead efforts to build the capacity of the branch. Coordinate membership support and training.

Departmental Representative for Experimental Psychology Oxford, UK **2023-Now**
University and College Union - University of Oxford Branch
Act as a point of contact for UCU members in the Department of Experimental Psychology.

Member of the Early Career Researcher Committee Oxford, UK **2023-2024**
Department of Experimental Psychology, University of Oxford
Represent ECRs in my department. Organize social events.

Symposium Organizer Online **2021**
Virtual Symposium on Explanation in Neuroscience and Artificial Intelligence
Organized one-day symposium to facilitate discussion among philosophers of science, neuroscientists and AI researchers. Also gave an introductory lecture and was a panelist during discussion panel.

Student Affairs Committee Member and Liaison to EDI Committee Montreal, Canada **2019-2020**
Unifying Neuroscience and Artificial Intelligence in Quebec (UNIQUE) research cluster
Represented student interests to the governance of UNIQUE and oversaw the organization of an annual student symposium

Workshop organizer Vancouver, Canada **2019**
NeurIPS workshop 'Real Neurons and Hidden Units: Future Directions at the Intersection of Artificial Intelligence and Neuroscience'
Orchestrated the open submission and review system and participated in all aspects of event organization

Reviewer **2014-Now**
Conferences: Computational and Systems Neuroscience (COSYNE), International Conference on Learning Representations (ICLR), Neural Information Processing Systems (NeurIPS) (rated in top 30% [2018] and top 10% [2020] of reviewers), Cognitive Computational Neuroscience (CCN), Women in Machine Learning.
Journals: Transactions in Machine Learning Research, Scientific Reports, Journal of Psychology and AI, European Journal of Neuroscience, Journal of Personality and Social Psychology, PLoS Biology, Current Biology, Discover Neuroscience, Hearing Research

Symposium organizer Hanover, USA **2013**
Dartmouth Arts and Music in Medicine and Neuroscience mini symposium
Made logistical arrangements, processed event registrations, advertised event

Hackathon organizer Montréal, Canada **2012**
Musitk
Secured sponsorship, processed reimbursements, oversaw logistics

SIG meeting organizer Hanover, USA **2012**
North East Music Informatics Special Interest Group (NEMISIG) meeting
Arranged catering, selected speakers, facilitated discussion sessions

Administrative Assistant Montréal, Canada **2010–2011**
Distributed Digital Music Archives and Libraries Lab, McGill University
Assisted with preparing large-scale grant applications

Equity, Diversity, and Inclusion

Secretary, Chair of the Records Committee **2015–2020**
Board of Directors, Women in Machine Learning Inc.
Led long-term organizational planning • Maintained all private records, the public WiML directory, all internal and outward-facing communication, websites and social media accounts • Trained and coordinated volunteers

Diversity and Inclusion Chair Montréal, Canada **2018**
Montreal Artificial Intelligence Symposium (MAIS)
Assisted the organizers to achieve their diversity and inclusion goals • Designed demographics questionnaire • Enforced and responded to violations of the code of conduct

Logistics Chair Montréal, Canada **2014**
Women in Machine Learning Workshop
Made local arrangements and participated in all aspects of event planning and fund raising